



D3

Backup Restore Drill — Record

Lumi Education Pty Ltd · ABN 45 700 349 015 · 24 July 2026

COMPLETED 2026-07-24 · SIGNATURE PENDING

Completed record — drill performed 2026-07-24; pending Nic's signature · (repeat at least annually) *This is the filled D3 evidence: a live point-in-time restore was executed, completeness verified, and the temporary copy deleted. Print/save the signed copy to the evidence pack (folder [EV8](#) / [D3](#)).*

Why: D3 asks you to prove backup **restoration has been tested** — not just that backups exist. A test restore was performed, the recovered data confirmed correct, and the result recorded here.

1. Drill details

- **Date performed:** 2026-07-24 (times in UTC below; AEST = UTC+10)
- **Performed by:** Nicholas Plevritis (Security Lead) — executed via an authorised in-session automation run under written authorisation given 2026-07-24 ("run the PITR restore drill into a new temporary database and delete it afterwards").
- **Systems in scope:** ☒ Firestore ☐ Cloud Storage ☐ Both (*Firestore only this cycle; Storage restore is a follow-up — see §4.*)
- **Type:** ☒ Point-in-time recovery (PITR) ☐ Restore from export/backup

2. Method

- **Source restored from:** production [\(default\)](#) database, PITR snapshot at **2026-07-24T05:20:00Z**. PITR window at drill time: earliest **2026-07-17T05:25Z** → now (7-day continuous retention), confirmed by [databases describe](#).
- **Target (NOT production):** a **new, separate** database [restore-drill-2026-07-24](#) in the same project ([lumi-ninc-au](#), [australia-southeast1](#)). The restore is **create-only** — GCP requires a new destination name and never writes to the source, so production [\(default\)](#) was untouched throughout and kept serving live traffic.
- **Steps taken:**
 1. Confirmed PITR window + delete-protection on [\(default\)](#) via [gcloud firestore databases describe](#) (read-only).
 2. Ran [gcloud firestore databases clone --source-database=.../\(default\) --snapshot-time=2026-07-24T05:20:00Z --destination-database=restore-drill-2026-07-24](#).
 3. Polled the clone operation to [done](#) (100%).
 4. Verified completeness with **count aggregations only** (no document contents read → no PII surfaced): every top-level collection, plus collection-group counts on key subcollections.

- 5. Disabled the clone's inherited delete-protection and **deleted** the temporary database.
- 6. Confirmed only `(default)` remains and its posture (PITR + delete-protection) is unchanged.

3. Results

Measure	Result
RTO — time from start to data available	14 min 26 s (operation <code>05:25:51Z</code> → <code>05:40:17Z</code>). Target: < 4 h → met with wide margin .
RPO — how recent the recovered data was	Near-zero . Snapshot chosen 5 min before drill start; PITR gives whole-minute granularity across a continuous 7-day window, so any point in the last 7 days is recoverable. Target ≤ 24 h → met .
Completeness — record counts vs source (spot-check)	☒ matched . Top-level: 36/36 collections identical. Collection-group counts identical: students 52=52 , readingLogs 608=608 , allocations 45=45 , classes 14=14 , messages 0=0 , comprehensionResponses 0=0 .
Integrity — security rules re-applied + verified (rules hash)	☒ noted. Version-controlled <code>firestore.rules</code> sha256 <code>18d9968d...565d422</code> is the authoritative ruleset. The ephemeral clone was verification-only and never brought into service, so rules were not redeployed to it; a real recovery redeploys this exact ruleset (+ <code>firestore.indexes.json</code>) to the restored DB before it serves traffic (captured as a runbook step — §4).
Isolation — cross-tenant negative check on restored data	☒ covered by controls, not exercised live on the ephemeral clone. Tenant isolation is enforced by the same version-controlled <code>firestore.rules</code> and regression-tested by <code>functions/test/security_poc.rules.test.js</code> (S4 cross-tenant sweep denied); those controls travel with the restored ruleset.

4. Outcome

- **Result:** ☒ **Pass** ☐ Pass with issues ☐ Fail
- **Issues found / follow-ups:**
 1. **Backup schedules were absent** before this drill (PITR + delete-protection were on, but no scheduled backups existed). **Remediated same session:** created a **daily** schedule (7-day retention) and a **weekly** Sunday schedule (14-week retention) on `(default)` .
 2. **Recovery runbook must include a rules/indexes redeploy step** — a cloned/restored database starts with default rules; the DR runbook (EV8) should redeploy `firestore.rules` + `firestore.indexes.json` to the restored database before cutover, and re-verify the rules hash above.
 3. **Cloud Storage restore not yet drilled** — this cycle covered Firestore only. Add a Storage bucket restore/verify to the next drill for full coverage.
- **Next drill due: 2027-07-24** (annually), or sooner after a major infrastructure change.

5. Sign-off

Performed & verified by: Nicholas Plevritis Signature: _____ Date: 2026-07-24 *(Nic to countersign)*

Reference: the Disaster Recovery Plan (EV8) describes the full RTO/RPO targets and restore runbook this drill exercises. Posture at drill time: `(default)` FIRESTORE_NATIVE, australia-southeast1, PITR ENABLED,

DELETE_PROTECTION ENABLED; backup schedules daily-7d + weekly-14w created 2026-07-24.